Name what is doing the pushing or pulling

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Interaction / ES: interacción / PT: interação

Every force has a source (what pushes or pulls) and a receiver.

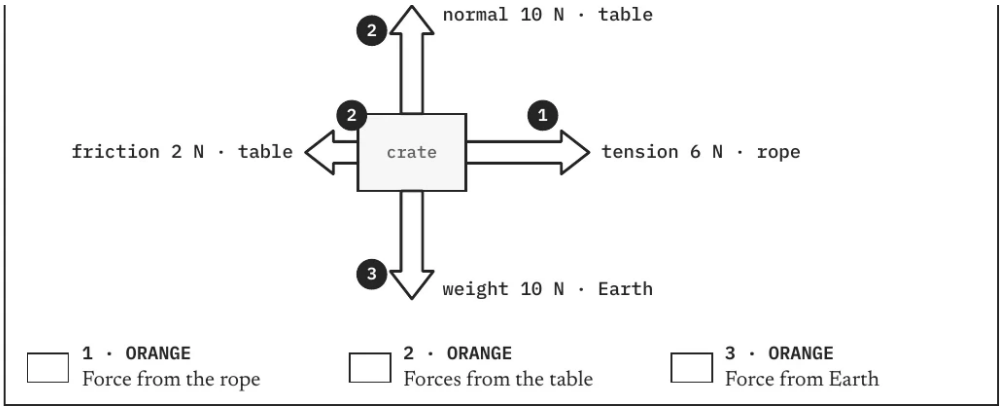
Also on your note page: Contact force, Non-contact force, Mass vs. weight.

B READ THE RELATIONSHIP

An interaction has two objects. One object exerts a force on the other.

Weight = mass × gravitational field strength

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg object rests on a level table. Find its weight and name the force that supports it.

Given m = 3 kg g ≈ 10 N/kg

Rule W = m × g

Work W = 3 × 10 = 30 N down table pushes up equally

Answer 30 N normal force 30 N up

E SAVE FOR THE FINAL MEETING

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

Read the physics: Interaction

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Ask: Which object receives the force? What other object pushes or pulls it? A rope pulls by contact. Gravity pulls even without contact.

02 HOW IT WORKS

Mass is measured in kilograms. Weight is a force in newtons. On Earth, multiply mass by gravitational field strength to find weight. Earth pulls downward; a table pushes perpendicular to its surface.

03 WHAT TO CHECK

A rope may carry part of a crate's weight. If the crate stays at rest, all upward forces together balance the downward force.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg object rests on a level table. Find its weight and name the force that supports it. 30 N normal force 30 N up.

05 CONNECT TO XJ

If XJ's path ever bends or speeds up, something is pulling on it. Naming the source of every force (out there, mostly gravity) is how analysts explain a change.

SAY IT BEFORE YOU WRITE

Who exerts the force, and on what object? The ___ exerts a force on the ___ toward ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Interaction

ES: interacción / PT: interação

Meaning in my words	How can you tell?
How would you explain interaction to a friend? In my words, this means ___.	In “the rope pulls the crate,” which two objects are interacting? I can tell because ___.
Example or sketch	**Non-example and why**
What interaction holds up a book resting on a table? My example shows ___.	Someone names “moving right” as the object pushing a cart. What would you ask? I would change ___ because ___.

Contact force

ES: fuerza de contacto / PT: força de contato

Meaning in my words	How can you tell?
How would you explain contact force to a friend? In my words, this means ___.	Which detail tells you the table’s push on a book is a contact force? I can tell because ___.
Example or sketch	**Non-example and why**
Draw the force a compressed spring exerts on a cart touching its right end. My example shows ___.	Earth pulls on a falling ball after it leaves your hand. Is that a contact force? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Non-contact force

ES: fuerza a distancia / PT: força à distância

Meaning in my words	How can you tell?
How would you explain non-contact force to a friend? In my words, this means ___.	A magnet tugs a paper clip before they touch. How can you tell contact is unnecessary? I can tell because ___.
Example or sketch	**Non-example and why**
Which downward arrow belongs on a ball after you let go? My example shows ___.	A hand holds up a book. Which force disappears when the hand moves away? I would change ___ because ___.

Mass vs. weight

ES: masa y peso / PT: massa e peso

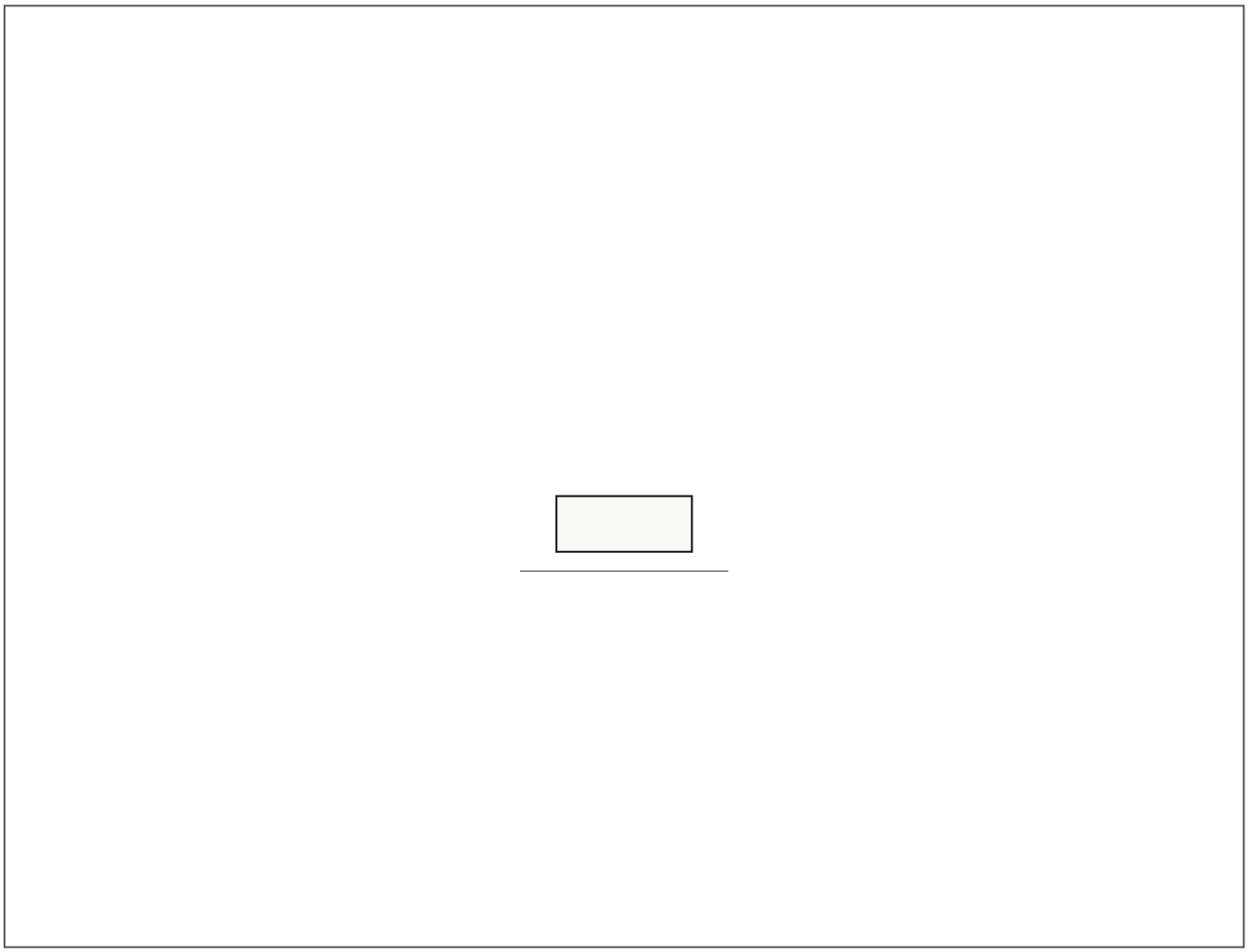
Meaning in my words	How can you tell?
How would you explain the difference between mass and weight? In my words, this means ___.	What changes when you take the same cart to the Moon: its mass or its weight? I can tell because ___.
Example or sketch	**Non-example and why**
On Earth, use 10 N/kg. What are the mass and weight of a 2 kg cart? My example shows ___.	A label says “mass = 20 N.” What does the unit tell you? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw Earth and a crate. Put weight on the crate; label source, receiver, direction, and N.

Labels you can use: source · receiver · contact.



Tell a partner

The ___ exerts a force on the ___ toward ___.

Who is doing the pushing?

10

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Name the source and receiver of each force.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

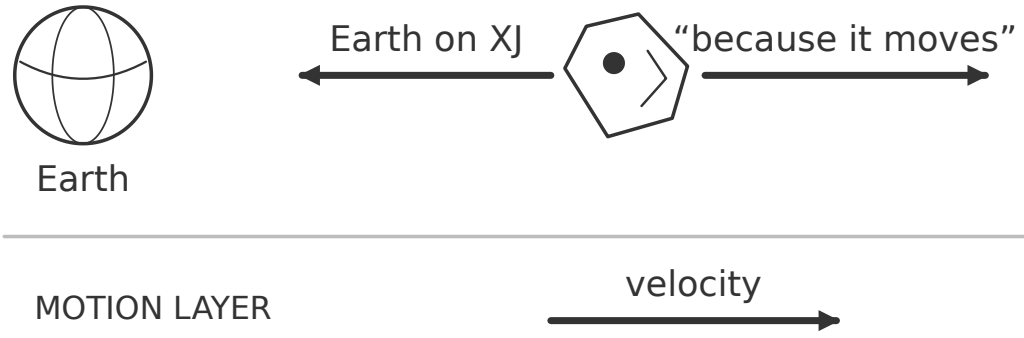
Approve the force layer for XJ. Name who exerts each force and what receives it. Keep motion separate.

SCIENCE GRAPHICS EDITOR ☐

Prepare the viewer graphic. Ask your partner who causes each proposed force before you approve its label.

Word help: source / receiver / gravity

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.



01 OUR FIRST CALL

Which force arrows should go live? Explain each arrow you keep or remove.

Keep ___ because ___ acts on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new information change the force layer? Explain your update.

Now we would say ___ because ___.

Switch speakers. Who causes each force you kept?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks